Jonathan Hawla

PhD student in Cosmology

CEA Paris-Saclay — IRFU / DPhP · DESI Collaboration · Université Paris-Saclay (ED 576 PHENIICS)

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Research Interests

Observational and computational cosmology: large-scale structure, Bayesian field-level inference with differentiable simulators, galaxy clustering × CMB lensing cross-correlations, primordial non-Gaussianity ($f_{\rm NL}$), GPU-accelerated inference.

Education

2025 – 2028 *(exp. Sep. 2028)*
PhD in Cosmology — CEA Paris-Saclay, IRFU/DPhP, Saclay, France
Université Paris-Saclay, Doctoral School ED 576 PHENIICS. Fully funded (*contrat doctoral*, 36 months).
Bayesian field-level inference with differentiable simulators for the joint analysis of DESI galaxy clustering and CMB lensing.
Supervisors: Étienne Burtin (director) & Arnaud De Mattia (co-supervisor).

2024 – 2025
M.Sc. (M2) Astronomy, Astrophysics & Space Engineering
Observatoire de Paris – PSL, Paris–Meudon, France. Graduated with honours (*mention bien*).

2023 – 2024
M.Sc. (M1) Universe Sciences & Space Technologies
Observatoire de Paris – PSL, Paris–Meudon, France. Graduated with honours (*mention bien*).

2020 – 2023
B.Sc. Physics
Université de Franche-Comté, UFR Sciences et Techniques, Besançon, France. Graduated with honours (*mention bien*).

Research Experience

Oct. 2025 – present
Doctoral research — CEA Paris-Saclay (IRFU/DPhP), Saclay, France

- Developed a fully differentiable, curved-sky forward model with a light cone (Lagrangian perturbation theory, galaxy bias, redshift-space distortions, Born-approximation lensing projector), in Python/JAX on GPUs at NERSC.
- Extended the field-level inference framework of H. Simon et al. (2025) to CMB lensing, building on `JaxPM` (W. Kabalan); posterior sampling with a Microcanonical Langevin Monte Carlo (MCLMC) sampler via `numpyro`.
- Identified and solved the *finite box* problem by analytically marginalising the out-of-box lensing contribution as a structured additive variance; implemented a pixel-exact likelihood enabling unbiased inference at small sky fractions.
- Validated the pipeline at realistic noise levels (DESI LRG shot noise, ACT DR6 reconstruction noise) on synthetic closure data and on AbacusSummit N-body simulations, quantifying the gain brought by CMB lensing on σ_8, galaxy bias and local primordial non-Gaussianity $f_{\rm NL}$.
- Implemented survey mask and footprint handling, enabling inference on realistic partial-sky geometries.
- Ongoing: model-misspecification studies and application to real data (DESI LRG × Planck PR4 / ACT DR6).

2025 (3 months)
Research internship — CEA Paris-Saclay (DPhP), Saclay, France
Supervisors: A. De Mattia & É. Burtin. Combined analysis of DESI 3D galaxy clustering and CMB lensing convergence maps from Planck and ACT.

2024 (2 months)
Research internship — LUTH, Observatoire de Paris, Meudon, France
Supervisor: P.-S. Corasaniti. Dark-energy models with a rapid phase transition in the context of the Hubble tension; Planck CMB likelihood analysis with `CLASS` and `MontePython` to constrain the *kink* parametrisation.

2023 (2 months)
Research internship — FEMTO-ST, Besançon, France
Supervisor: J. Gillot. Impact of optical power fluctuations on the stability of a laser stabilised by a cryogenic Fabry–Pérot cavity, in preparation for a power-servo loop.

2025 (1 week)
Observing run — Observatoire de Haute-Provence, France
Spectroscopy of Be stars with the T152 telescope: observation planning, telescope operation, data reduction.

2025
Methodology project — Observatoire de Paris, Meudon, France
Supervisor: Y. Rasera. Particle-Mesh N-body simulation of galaxy clustering implemented in Fortran; team project with Git-based version control.

Publications

No refereed publication yet. First-author paper in preparation on field-level inference for the joint analysis of DESI galaxy clustering and CMB lensing.

Talks

Nov. 2026
Galaxy clustering × CMB lensing at the field level
Action Dark Energy 2026, Paris, France — contributed talk.

Jul. 2026
Adding CMB lensing to the galaxy clustering field-level analysis
DESI Collaboration Meeting, Durham, UK — Primordial Non-Gaussianity working group, contributed talk.

Jun. 2026
Field-level inference for the joint analysis of galaxy clustering and CMB lensing
GDR CoPhy annual meeting, Clermont-Ferrand, France — contributed talk.

Apr. 2026
Field-level inference for the joint analysis of galaxy clustering and CMB lensing
Elbereth Conference, Institut d'Astrophysique de Paris, Paris, France — contributed talk.

Dec. 2025
Spotlight talk — DESI Collaboration Meeting, Tucson, AZ, USA.

Posters

Jun. 2026
Field-level inference for the joint analysis of galaxy clustering & CMB lensing
PheniicsFest, annual meeting of the ED PHENIICS, Université Paris-Saclay, Orsay, France.

Schools & Training

Aug. 2026
Rodolphe Clédassou (Euclid) Summer School — Cycle 2 (Theory), 2 weeks, Pornichet, France.

Aug. 2025
Rodolphe Clédassou (Euclid) Summer School — Cycle 1, 2 weeks, Les Karellis, France.

2025 – 2026
Doctoral training — Research ethics and scientific integrity (MOOC); prevention of gender-based and sexual violence and of discrimination. Université Paris-Saclay.

Collaboration Service

2025 – present — **DESI Collaboration** — active member; contributor to the Primordial Non-Gaussianity working group. Three-night observing shift as remote Supporting Observer.

Teaching & Outreach

2024 – present — **Private tutoring** — high-school mathematics (3 h/week) and piano (1 h/week).

2022 — **Volunteer organiser** — SF2A annual meeting (French Society of Astronomy & Astrophysics), 1 week, Besançon, France.

Software & Projects

2025 – present — `thesis-desi-cmb-fli` — differentiable forward model and Bayesian inference pipeline for the joint field-level analysis of galaxy clustering and CMB lensing (Python/JAX, multi-GPU). Open source (MIT), with unit tests, continuous integration and documentation.

2022 – 2026 — **CarbonScale** — designer and developer of an app to track and reduce personal CO_2 emissions. Winner of the *Pépite Bourgogne-Franche-Comté* idea competition (2023). `carbonscale.app`

Technical Skills

Languages	Python (NumPy, JAX, `numpyro`, Astropy, HEALPy), Fortran, Bash, LaTeX.
Cosmology	`CLASS`, `MontePython`, `JaxPM`, `HEALPix`, AbacusSummit simulations.
Methods	Bayesian inference, MCMC/HMC/MCLMC samplers, automatic differentiation, GPU computing (NERSC), N-body and LPT simulations.
Tools	Git, Linux/HPC environments, Keynote, Lightroom.

Languages

French — native speaker. **English** — fluent; TOEIC 965/990 (CEFR C1, 2023), EF SET 72/100 (C2, 2023); all talks given in English. **Lebanese Arabic** — fluent spoken (not written).

Interests

Piano (ten years at the conservatoire, band *Puls'art*) — first-aid volunteer, French Red Cross (PSE2) — Muay Thai — photography — chess — philosophy of science.

References

Étienne Burtin (etienne.burtin@cea.fr) and **Arnaud De Mattia** (arnaud.de-mattia@cea.fr), CEA Paris-Saclay, IRFU/DPhP.